

Employment

DPhil Researcher, University of Oxford

Oct '25 - present

- Full scholarship for the [Intelligent Earth: UKRI CDT in AI for the Environment](#) doctoral training programme.
- Research with [Milan Klöwer](#) and [Laura Mansfield](#) developing generalisable hybrid physics + AI parameterisations for [SpeedyWeather.jl](#), a Julia-based global atmospheric model. Focuses on online learning, physical constraints and generalisability in space, under extremes and under climate change.
- Continued development of [ProfSea-Climate](#), contributing simulations to the [sea-level emulator intercomparison project](#) (in prep.) and developing statistical emulations of physical models.

Applied Climate Scientist, Project Scientist, Met Office

Feb '22 - Oct '25

- Lead author of a [high-impact paper](#) presenting impacts of overshoot on vulnerable ecosystems for new policy-relevant emissions pathways (Nature Climate Change, 2025).
- Co-author of a technical paper presenting a [novel emissions-to-impacts climate emulator system](#) (GMD, 2025).
- Lead developer of the Met Office [sea-level projections emulator tool](#) (in prep.).
- Led the analysis for a [flood modelling climate attribution study](#) funded by the COMPASS EU project (COMPASS, 2025).
- Developed ML time series and vision algorithms to classify and predict lightning types and strengths; used HPC GPU clusters to train deep NN architectures.
- Developed a U-Net deep convolutional model for computationally efficient temporal downscaling of climate model output fields, training and inference on HPC GPU clusters.
- Developed a climate data and patterns processing pipeline for an [IPCC-endorsed climate model evaluation tool](#).
- Worked with project managers and colleagues to successfully deliver on project deadlines, taking on additional strategically important work and responsibilities beyond my job description; including task managing and mentoring several junior colleagues.
- Presented work as a scientific poster and gave talks at international science conferences *AGU 2022*, *AGU 2023* and *EGU 2024*; convened a climate emulators session at *EGU 2025*; ran tutorial sessions to a large group of senior colleagues/managers on an Earth System Model evaluation tool; gave numerous internal talks about project work.
- Converted a [complex algorithm written in archaic Fortran](#) to Python, enhancing the capabilities and value of an experimental Earth observations network.

Product Lead at i24

Mar '21 - Jan '22

- Designed and helped build a FinTech start-up mobile app.
- Pushed front-end development updates through self-taught React.js and React Native.
- Designed and co-wrote the company pitch deck (calculating financial projections, solidifying revenue model and unit economics).
- Derived business development insights, analytics and forecasts from growth and user data.

Student Ambassador for the University of Exeter

Jan '19 - Aug '20

- Representing the University to prospective students, working especially in outreach programs for prospective students from low-income backgrounds.
- Developing leadership and communication skills, and customer interfacing experience.

Education

DPhil Intelligent Earth (UKRI CDT in AI for the Environment), University of Oxford

2025 - present

Master of Physics with Honours in Physics with Astrophysics, University of Exeter

2016-20

A-Levels, Abingdon School

2014-16

Physics (A), Mathematics (A), Chemistry (A) and French AS (A).

Further Experience/Skills

Internships

- Fluid Dynamics at the University of Oxford Thermofluids Institute (3 months), high-school internships at the University of Reading Meteorology dept. and Diamond Light Source (particle accelerator).

Creative Arts

- Grade 8 Voice, grade 6 Piano, and grade 4 Traditional Irish Violin (all distinction).
- Freelance photographer for hospitality marketing, blogs and university societies.

Sports

- Avid surfer regularly organising groups of unorganisable surfers on weekday, weekend, and international trips.
- Further sporting interests include climbing, snowboarding, mountain biking and running.